

Polygenic Risk Scores

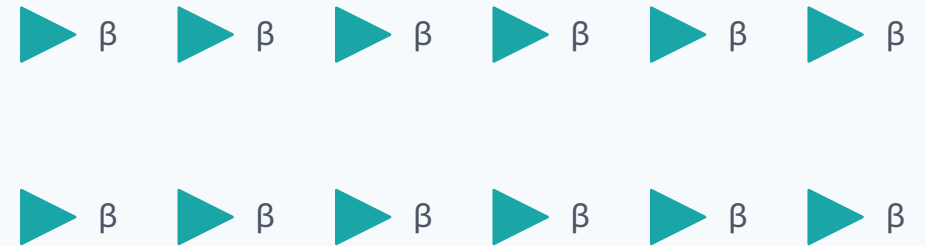
From GWAS signals to individual-level risk prediction

- Define a PRS and how it is constructed from GWAS results
- Understand key modeling choices (LD, SNP selection, shrinkage)
- Interpret performance metrics and what PRS can/can't tell you
- See real examples (CAD, breast cancer, BMI) and major limitations (portability)

Many common traits & diseases are highly polygenic

- Individual GWAS hits usually have small effects → limited predictive power alone
- Aggregating many SNPs can capture a meaningful fraction of common-variant liability
- A PRS is a single number you can compute once and use across the life course
- Most useful for risk stratification (percentiles), not deterministic diagnosis

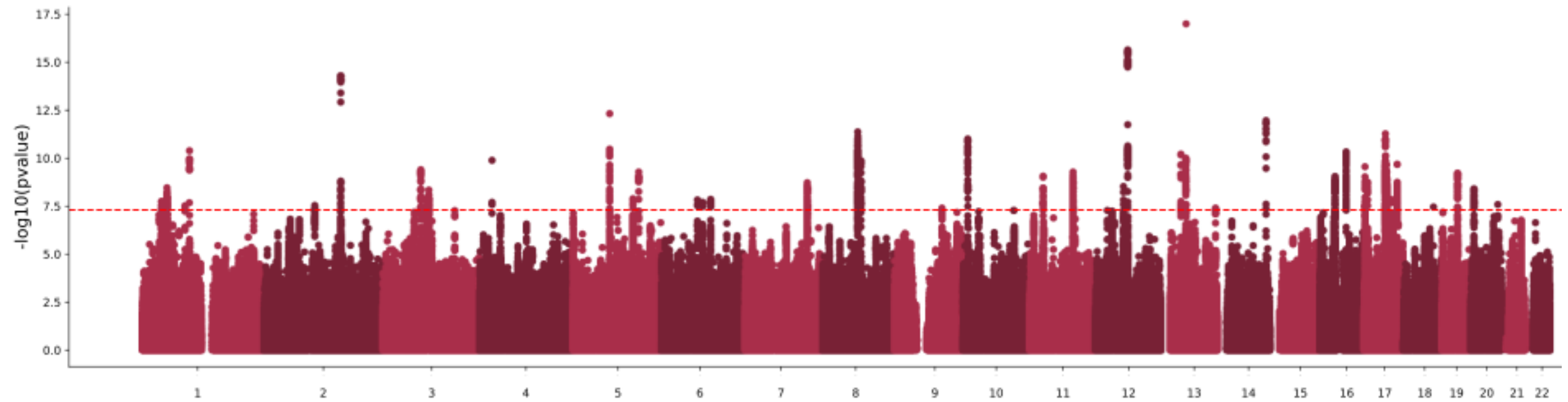
Polygenic architecture (conceptual)



Sum of many
small effects

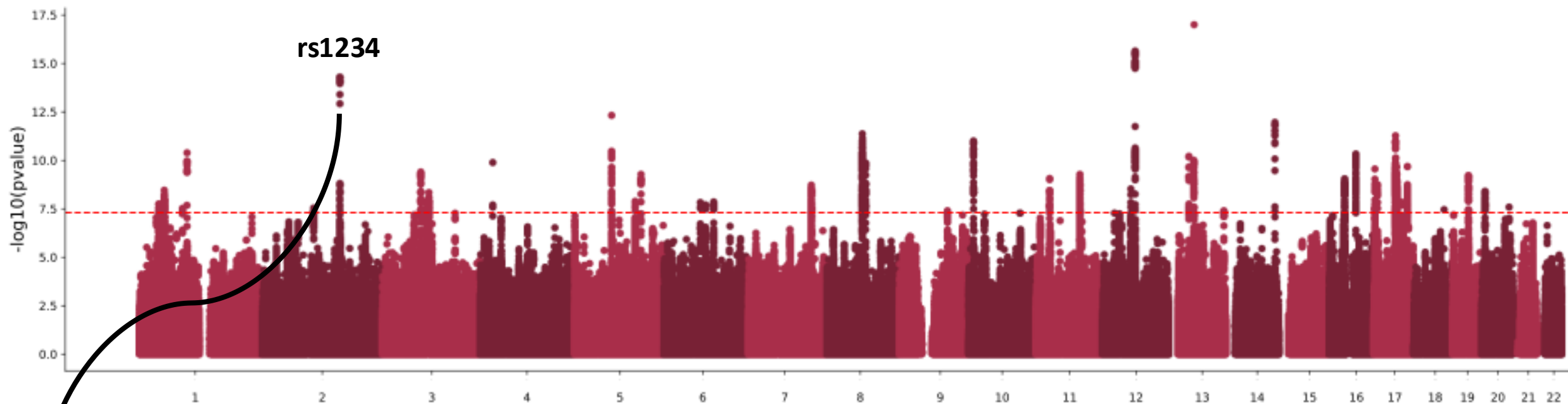
PRS → risk percentile

Polygenic risk score calculation illustration

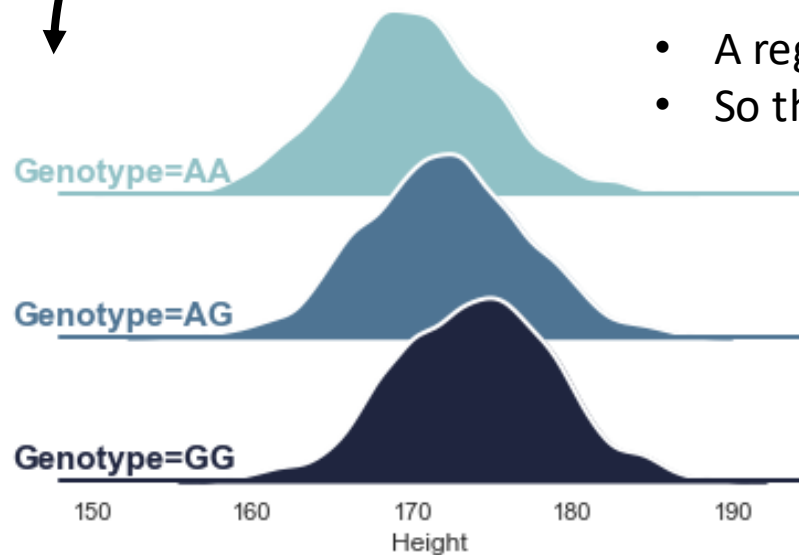


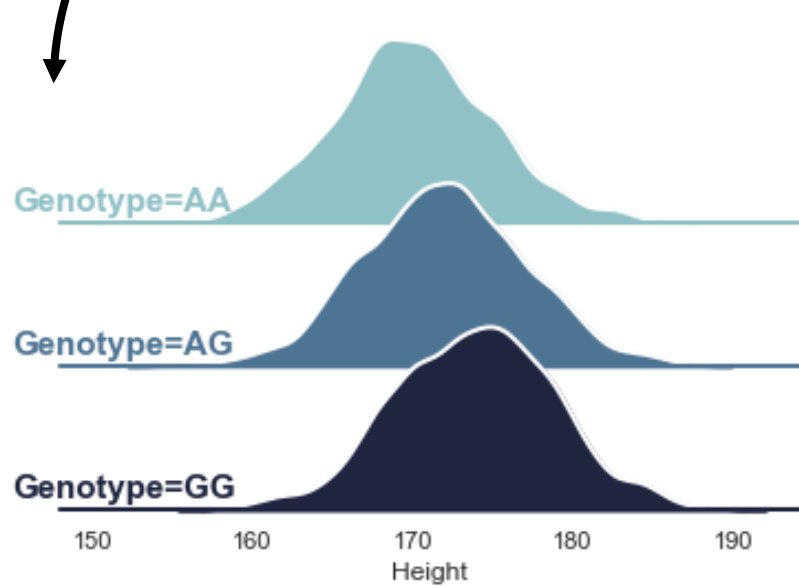
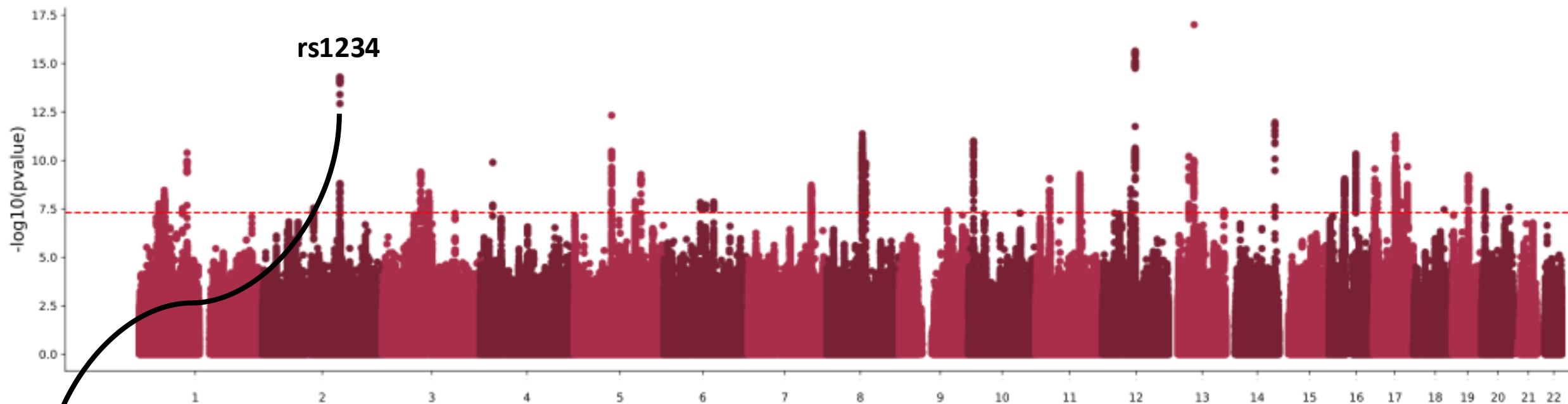
Genome-wide association study of height

GWAS of height



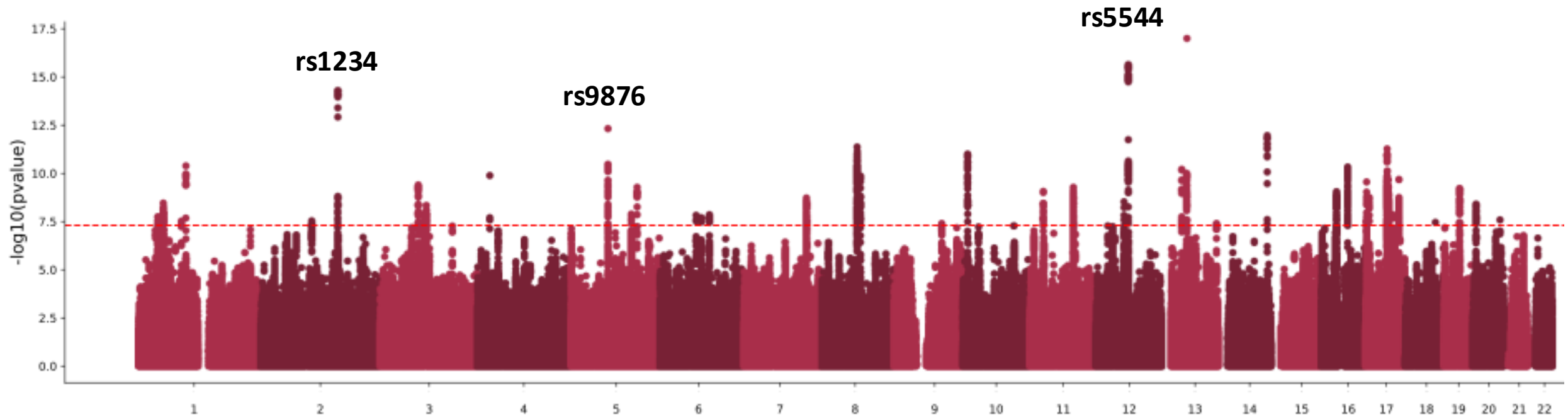
- A regression would show an average increase of 2cm per copy of the G allele at rs1234.
- So the effect size (beta coefficient) of this variant would be approximately 2.





In a new sample we would expect AG individuals at rs1234 to be on average 2cm taller than AA and 2cm shorter than GG.





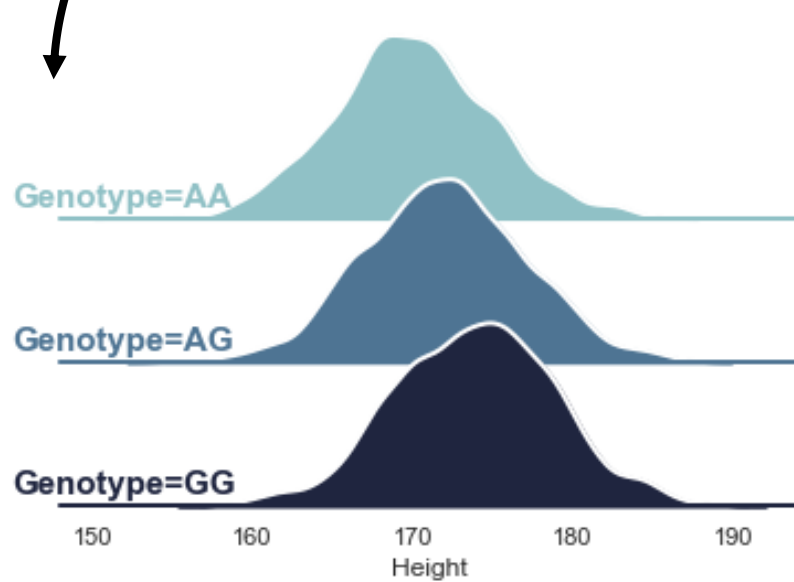
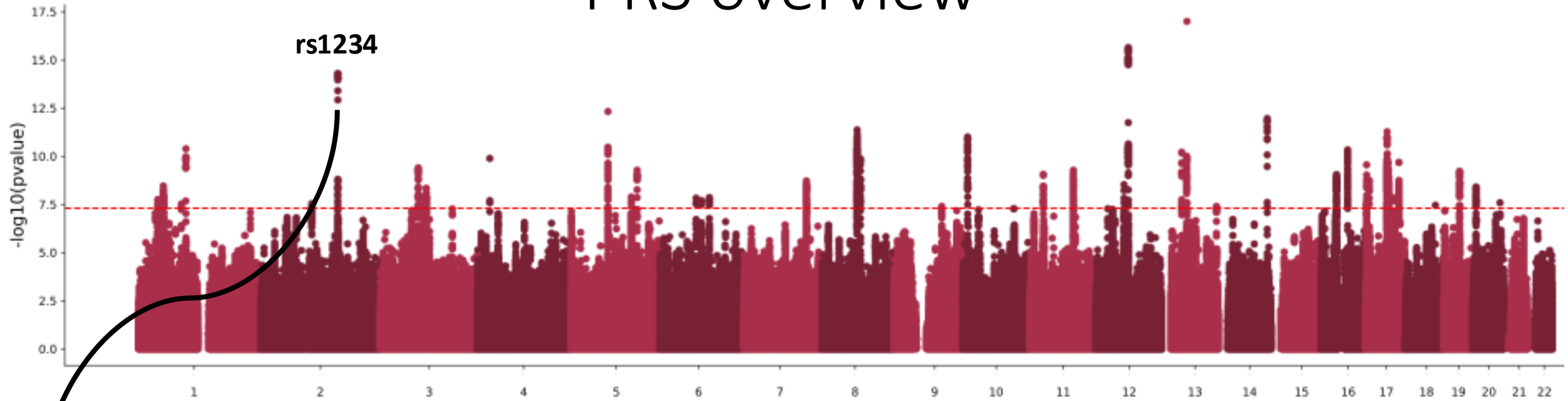
Complex traits are highly polygenic!

From above we can see there are many more genetic variants that contribute to the phenotype

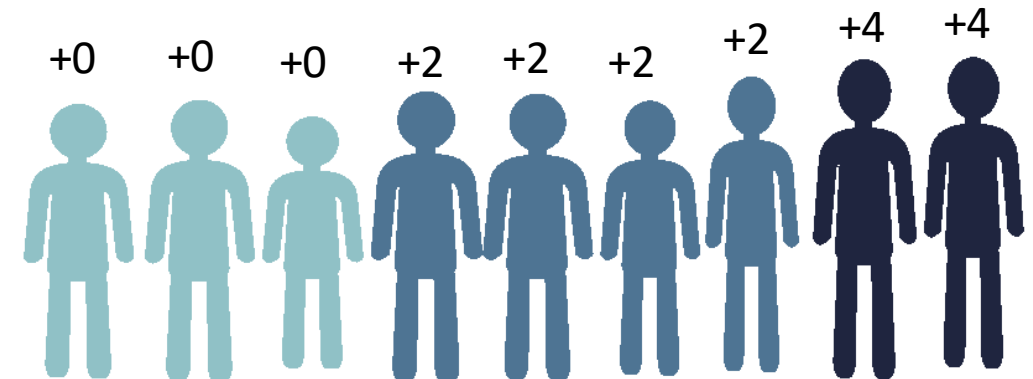
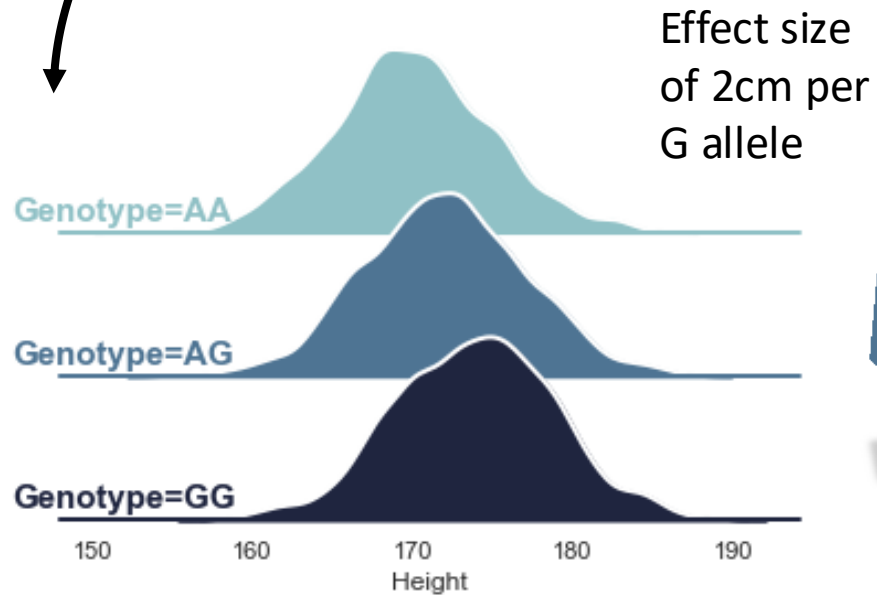
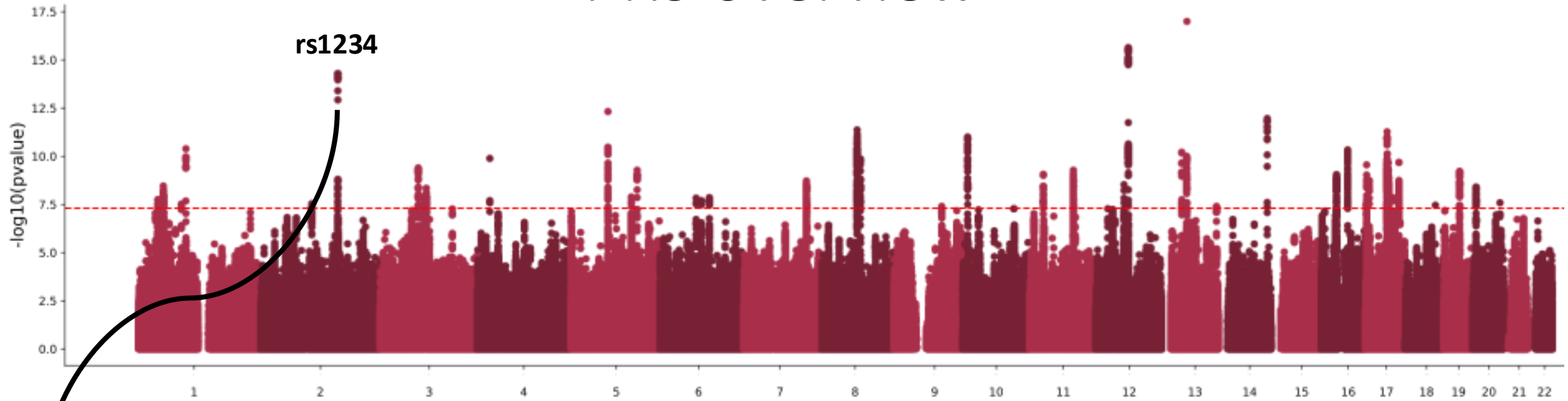
Common variants typically have a small effect size (our example is an exaggeration for a common variant). This would cause single-loci based prediction useless

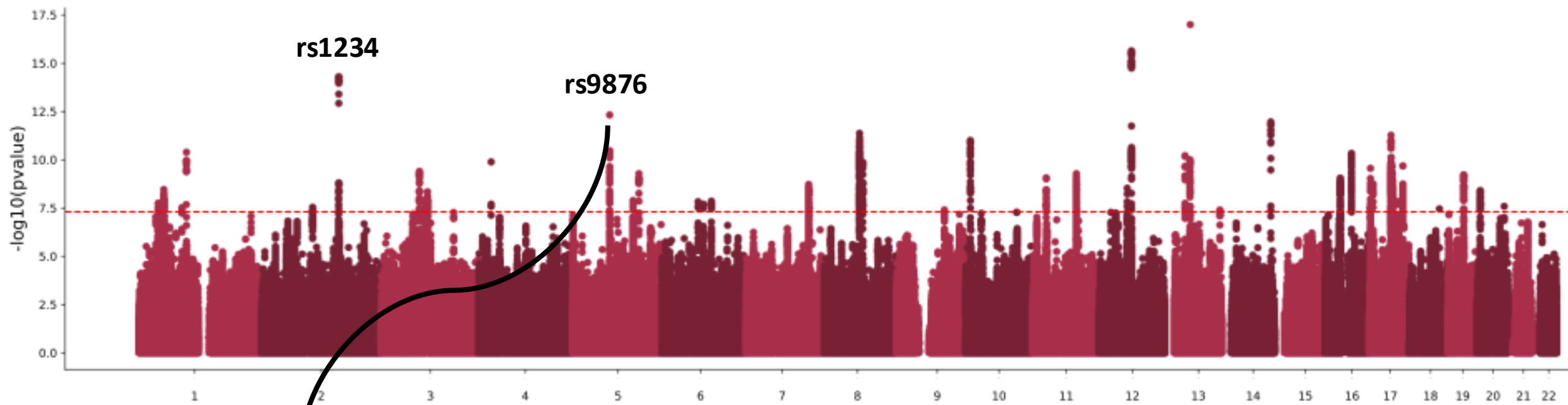
We can combine the information we gain from several genetic variants to estimate an overall score and gain a better estimate of the trait. This is essentially what a PRS does.

PRS overview

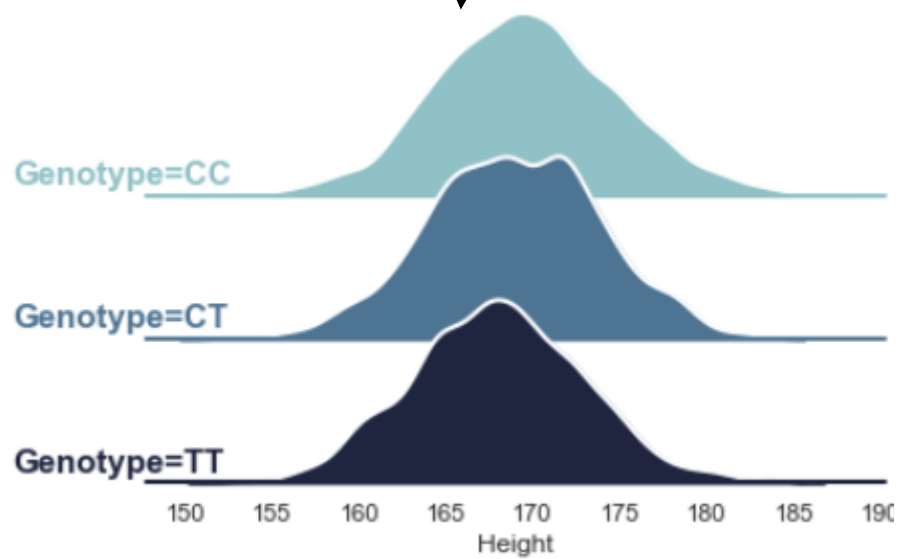


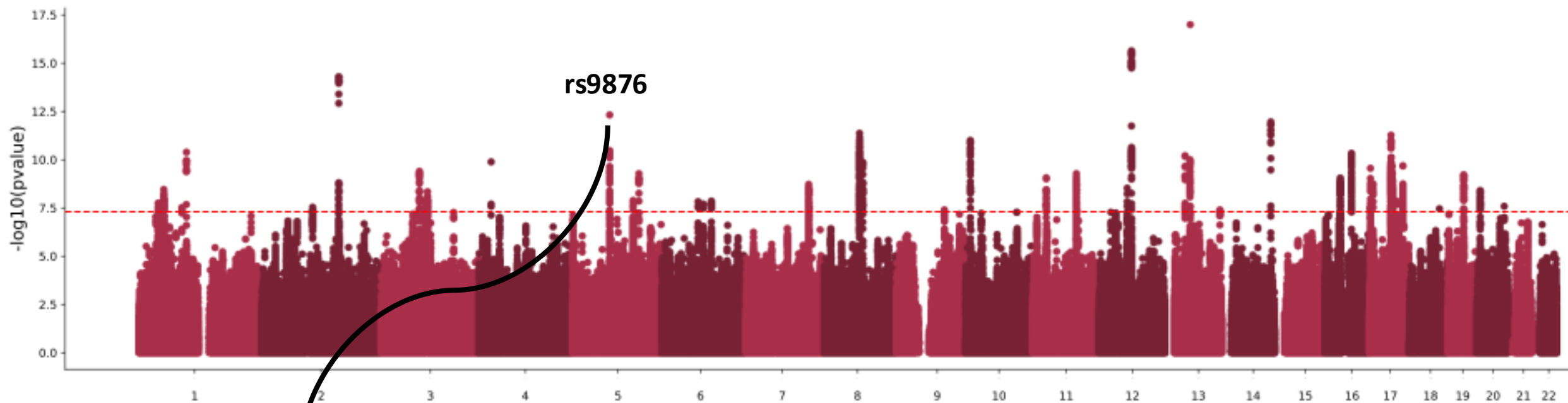
PRS overview



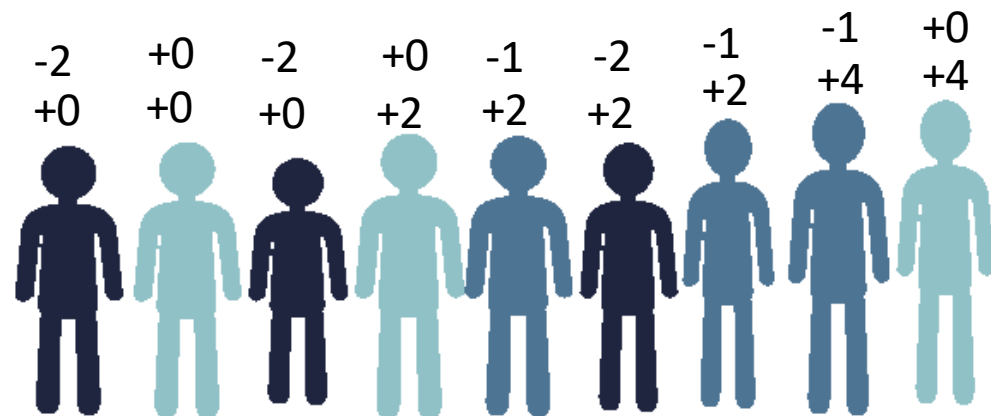
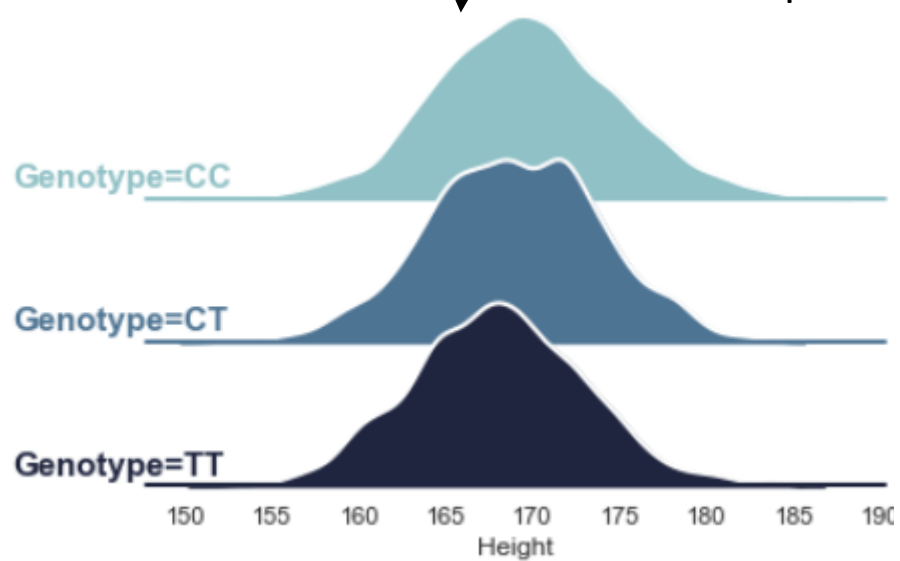


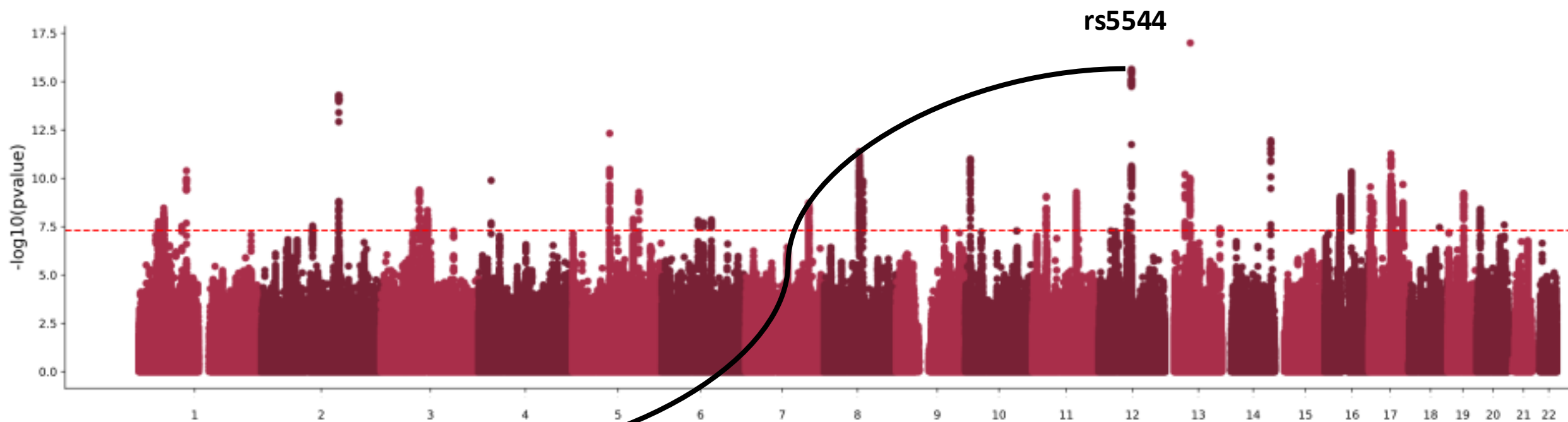
Effect size of -1 per T allele



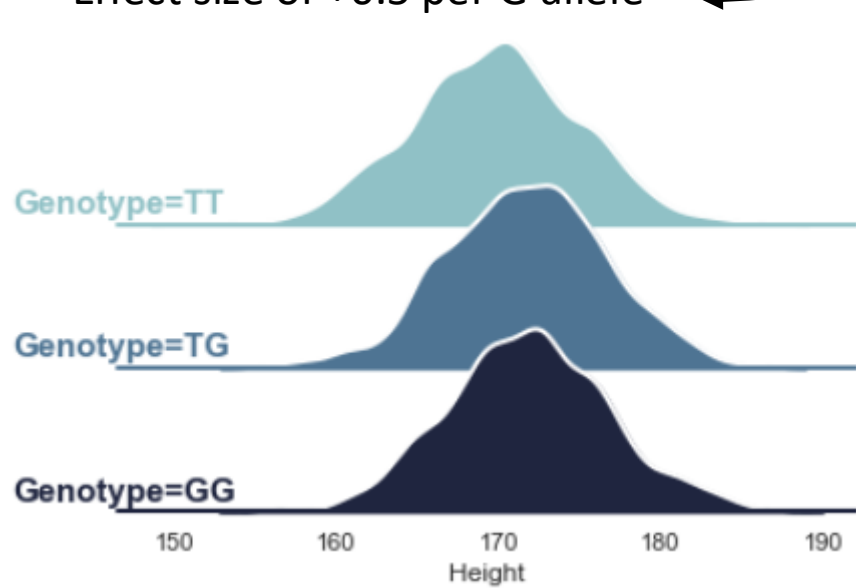


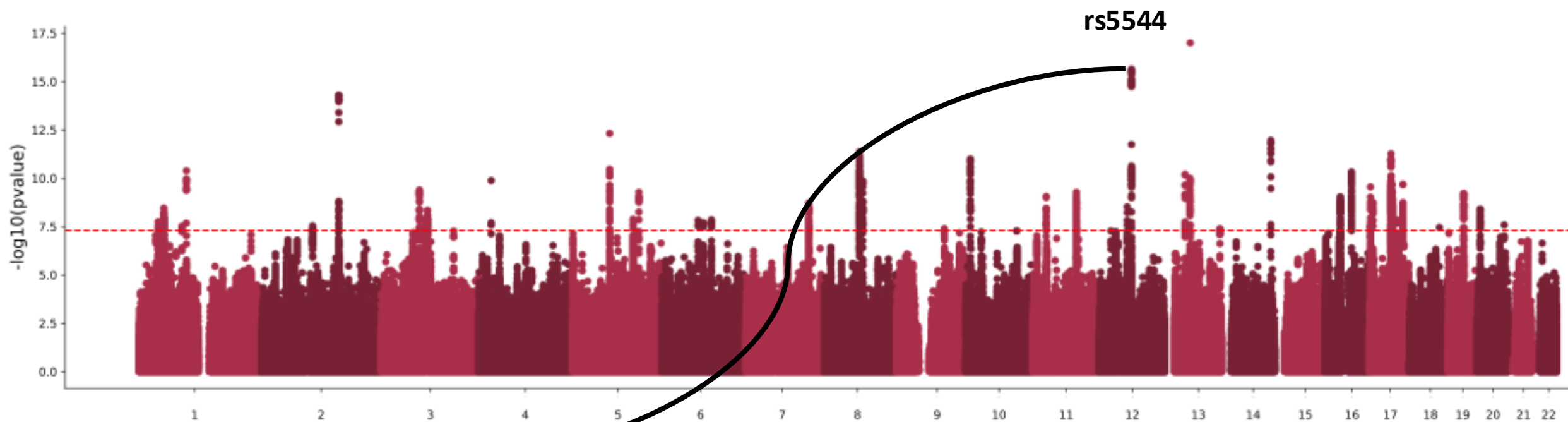
Effect size of -1 per T allele



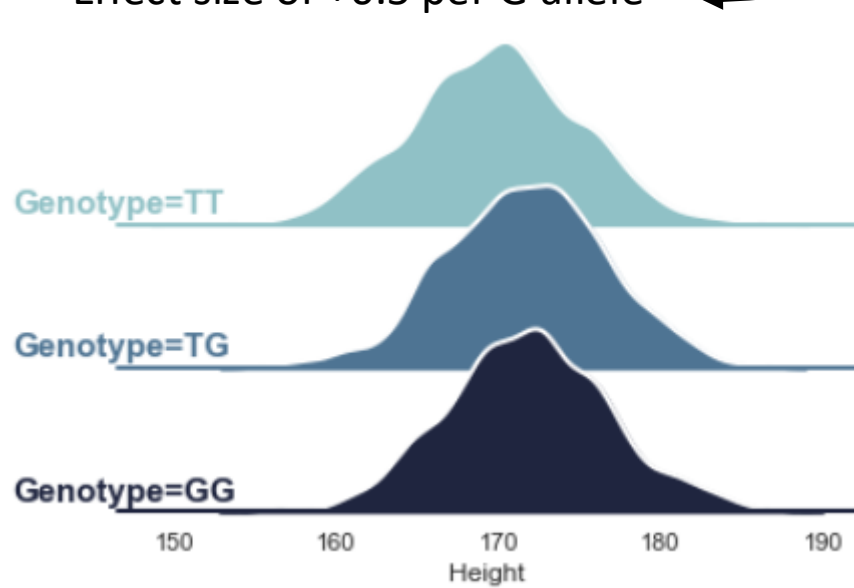


Effect size of +0.5 per G allele

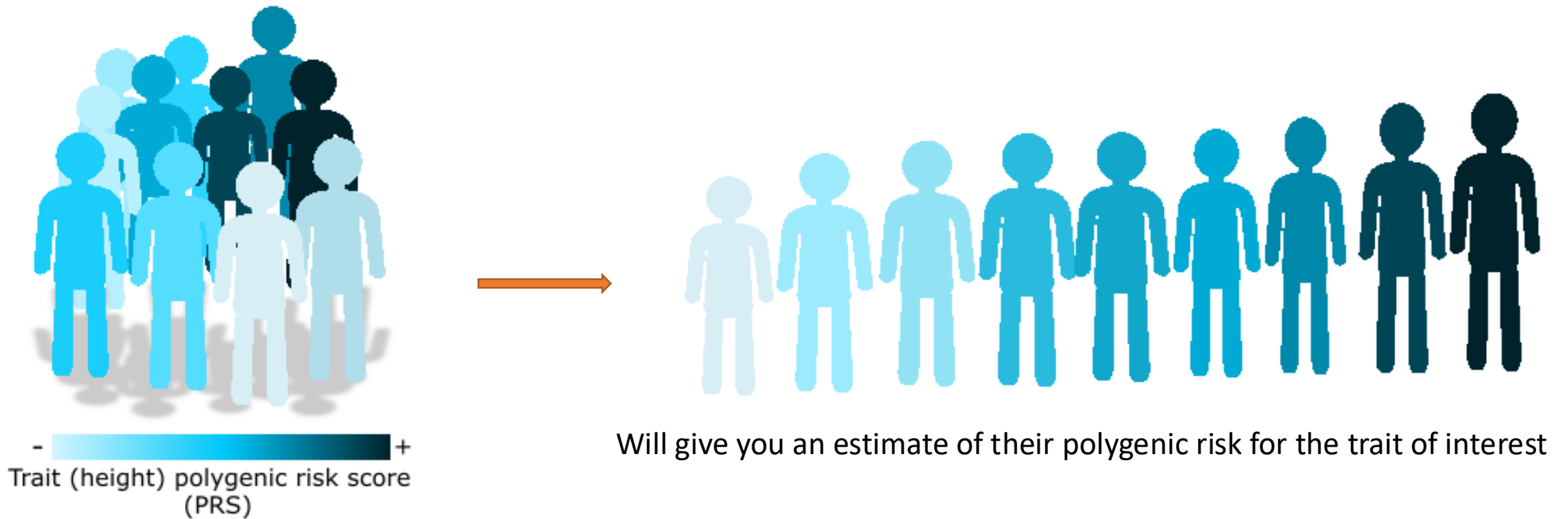




Effect size of +0.5 per G allele

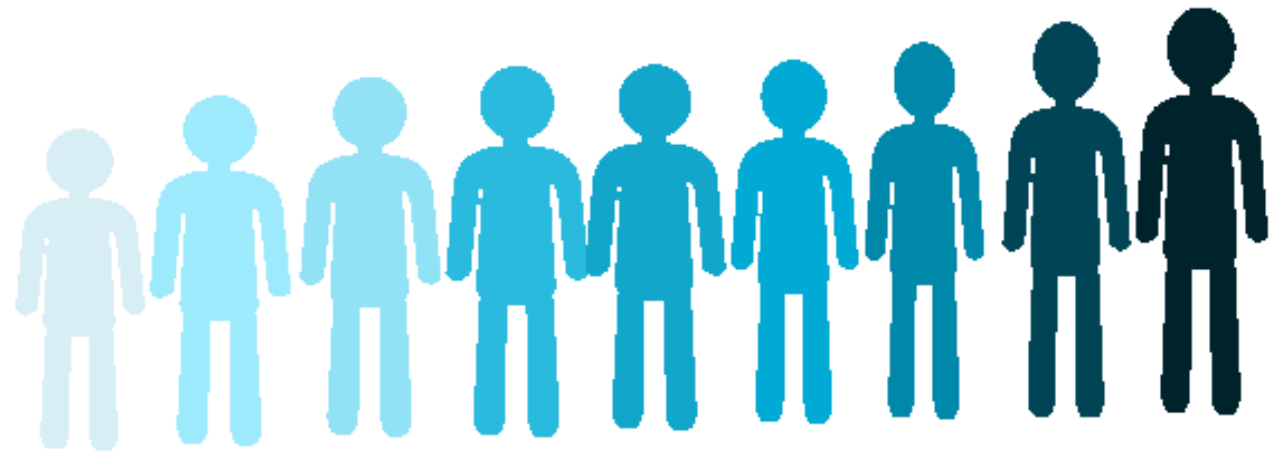


Repeat including the other variants and sum across all loci



Polygenic risk score – Weighted sum of alleles which quantify the effect of several genetic variants on an individual's phenotype.

Repeat including the other variants and sum across all loci



Caution! The sample for which PRS will be calculated should be independent from that of the discovery GWAS. Sample overlap will bias your results.

Definition

A polygenic risk score is a weighted sum of an individual's genotypes across many variants.

$$PRS_i = \sum_j (G_{ij} \times \hat{w}_j)$$

G_{ij} : genotype dosage (0/1/2) for individual i at SNP j

$\hat{w}_j$: weight from a discovery GWAS (e.g., β or $\log(\text{OR})$ for case/control)

Important details

- Harmonize alleles between GWAS summary stats and target genotypes (A/T and C/G need care)
- LD means SNP effects are correlated $\rightarrow$ naive “all SNPs” can over-count signal
- Scores are often standardized (mean 0, SD 1) within the target population

From GWAS to PRS: two datasets

PRS analyses are defined by:

- Base (discovery) data: GWAS summary statistics
- Target data: genotypes + phenotype (for validation or application)
- Keep base and target independent to avoid inflation from sample overlap
- Use an LD reference panel that matches the target ancestry when possible

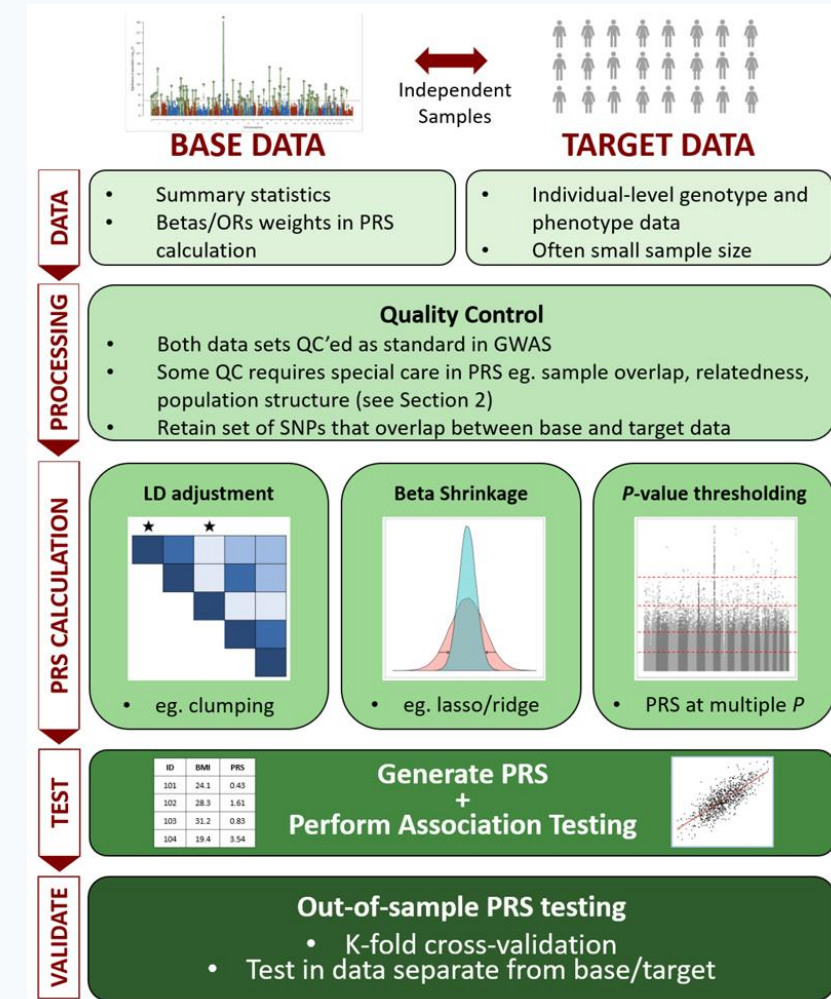
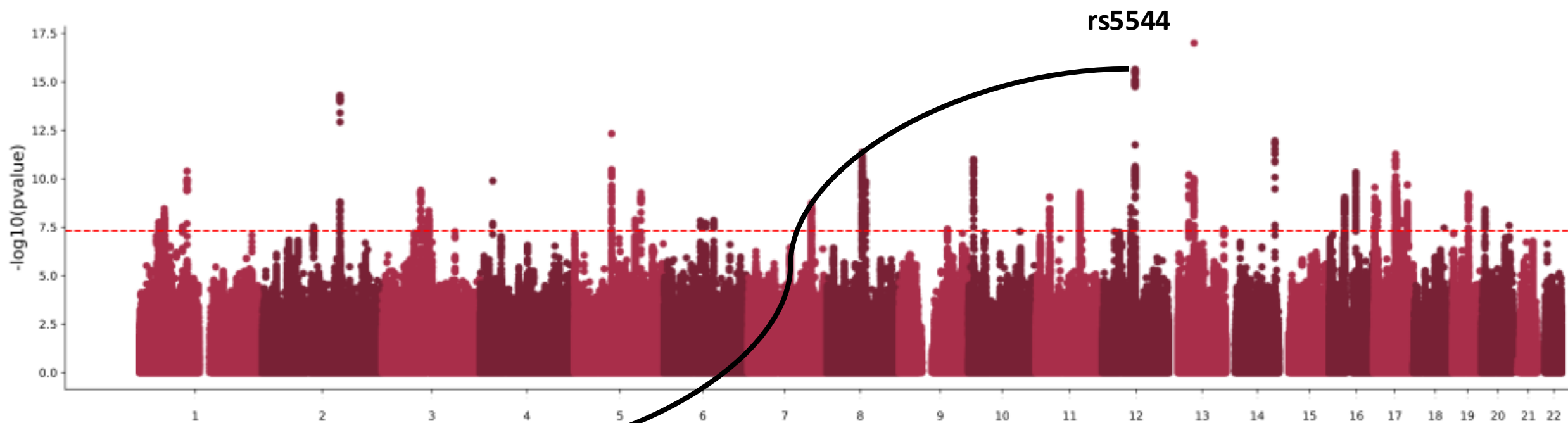
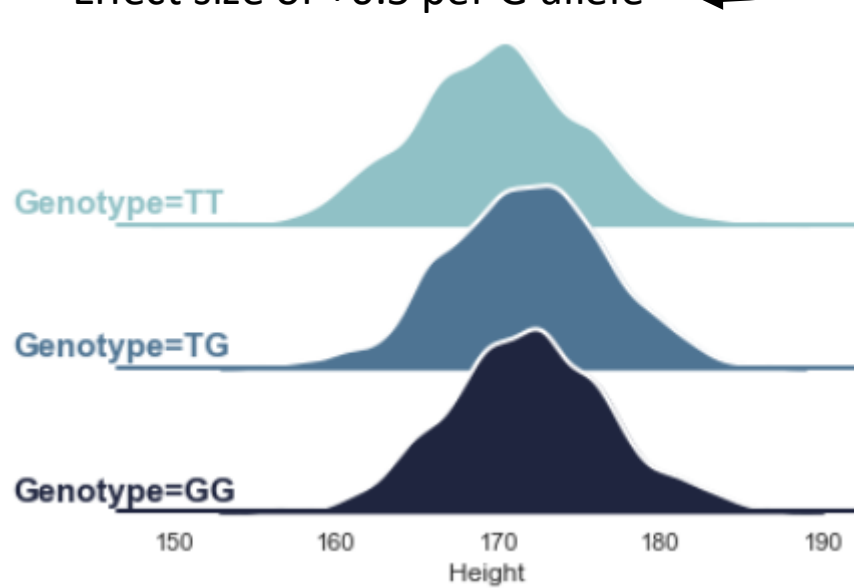


Figure: PRS analysis process (base vs target, QC, LD, shrinkage)



Effect size of +0.5 per G allele



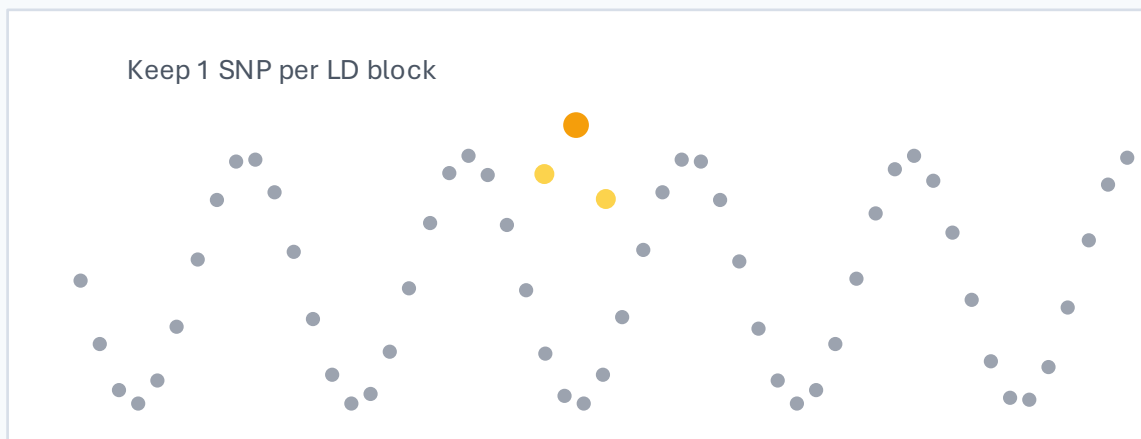
A baseline method: clumping + thresholding (C+T)

Idea

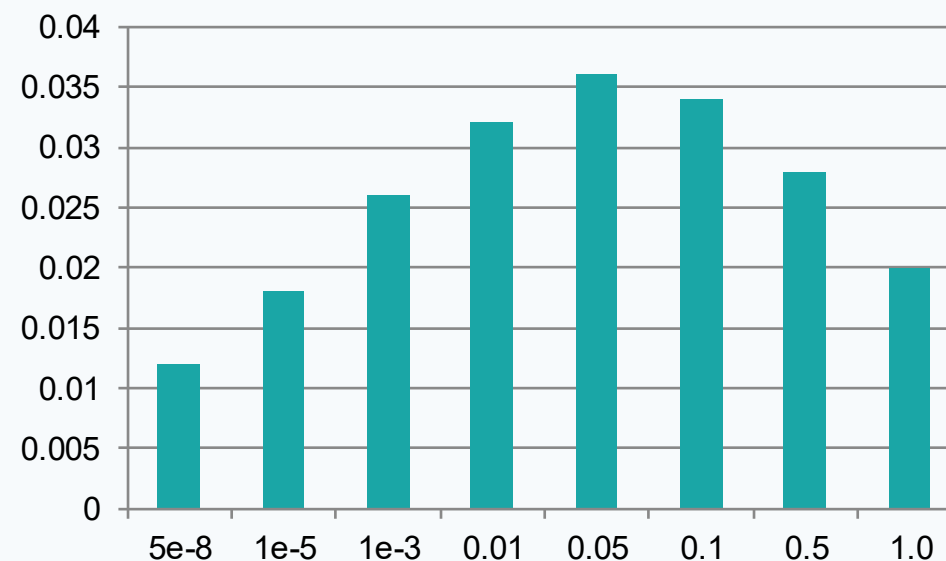
- Threshold: keep SNPs with GWAS p-value $< P_t$
- Clump: within LD windows, keep the most significant SNP
- Try multiple P_t values and pick the best in a validation set

LD clumping (conceptual)

Keep 1 SNP per LD block



Tuning P_t : pick the best threshold in validation

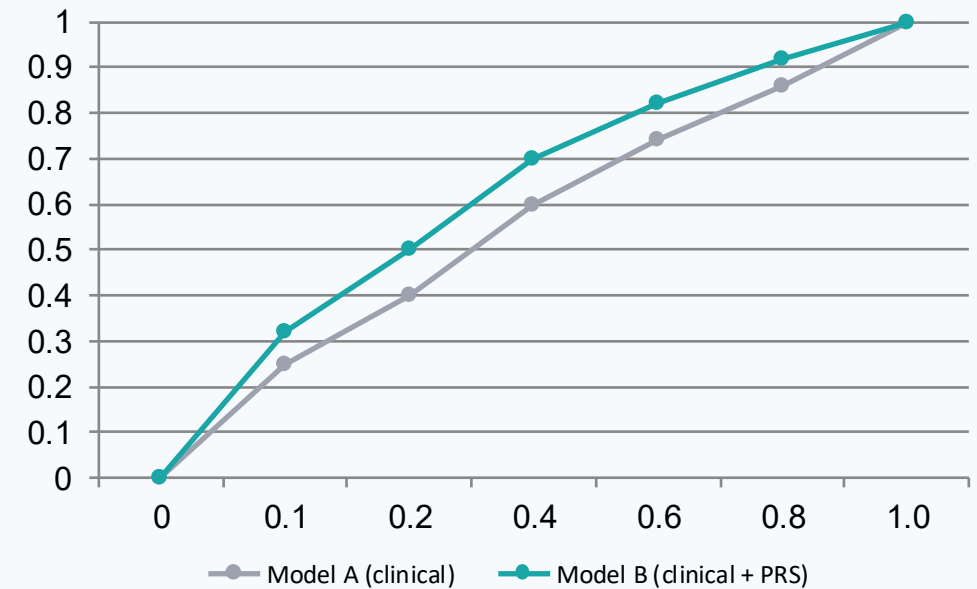


PRS version	p-value threshold	R ² in validation set
PRS ₁	p < 5e-8	0.8%
PRS ₂	p < 1e-5	1.4%
PRS ₃	p < 0.01	2.1% ← best
PRS ₄	p < 0.1	2.0%

Common metrics

- Quantitative traits: R^2 , correlation, mean-squared error
- Binary traits: AUC/C-index, odds ratio per SD, liability-scale R^2
- Net benefit: does PRS improve decisions vs existing models?

ROC curve (illustration)

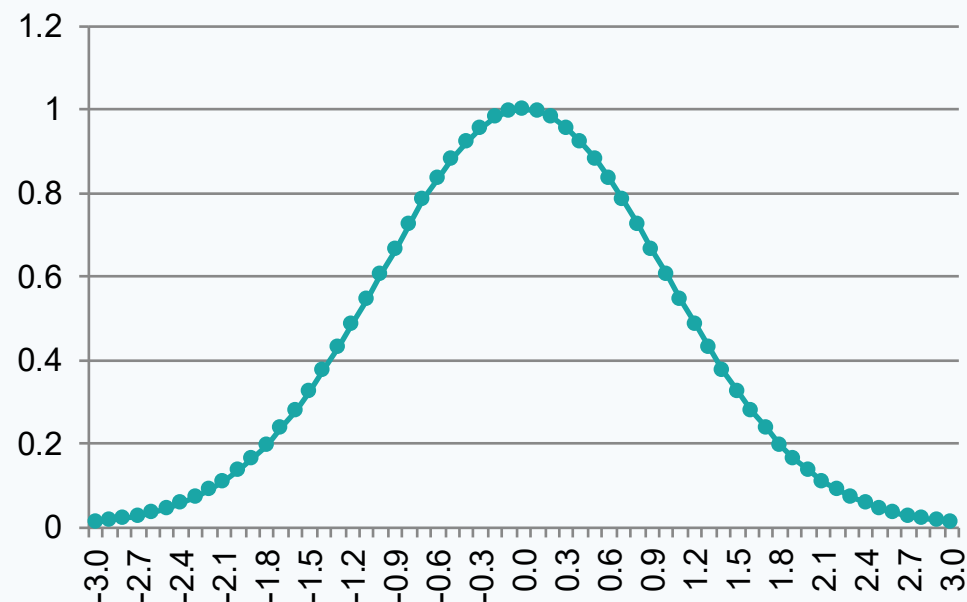


AUC summarizes discrimination

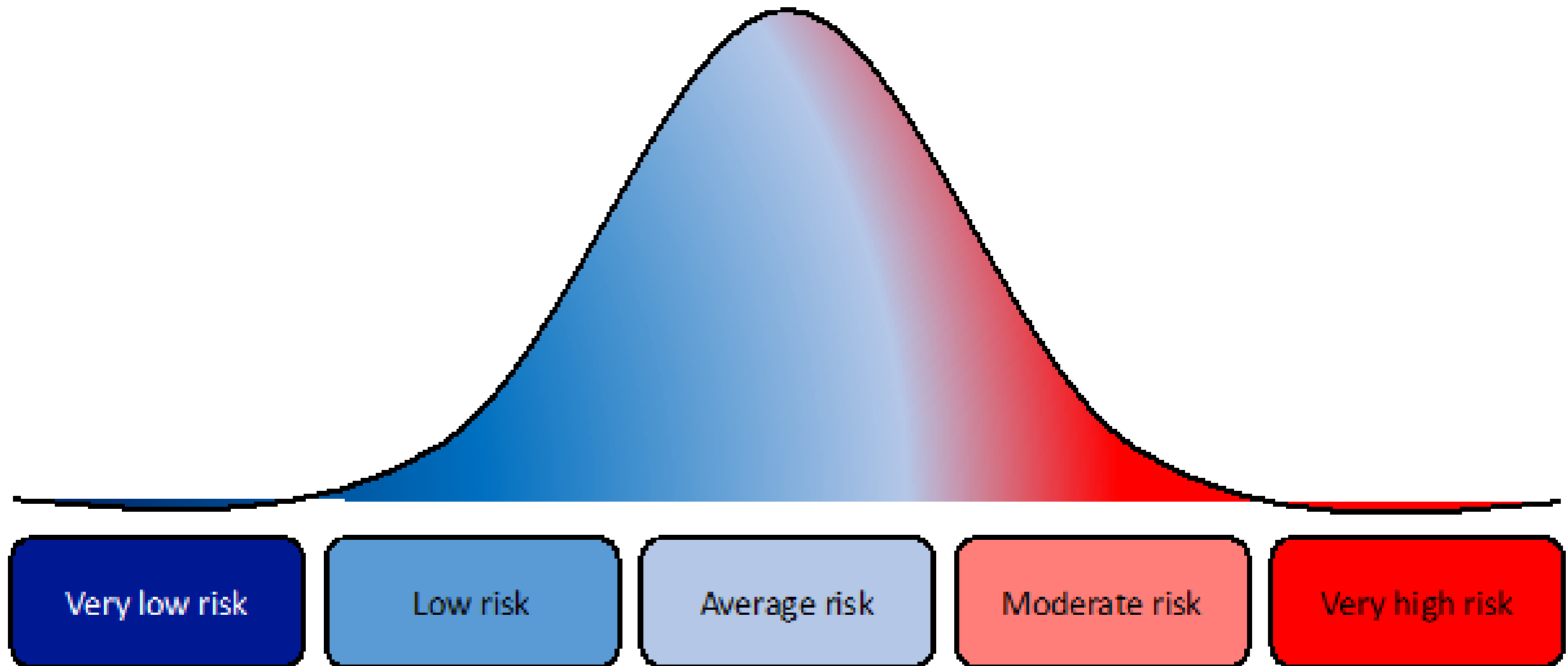
PRS values are (approximately) normally distributed within an ancestry group

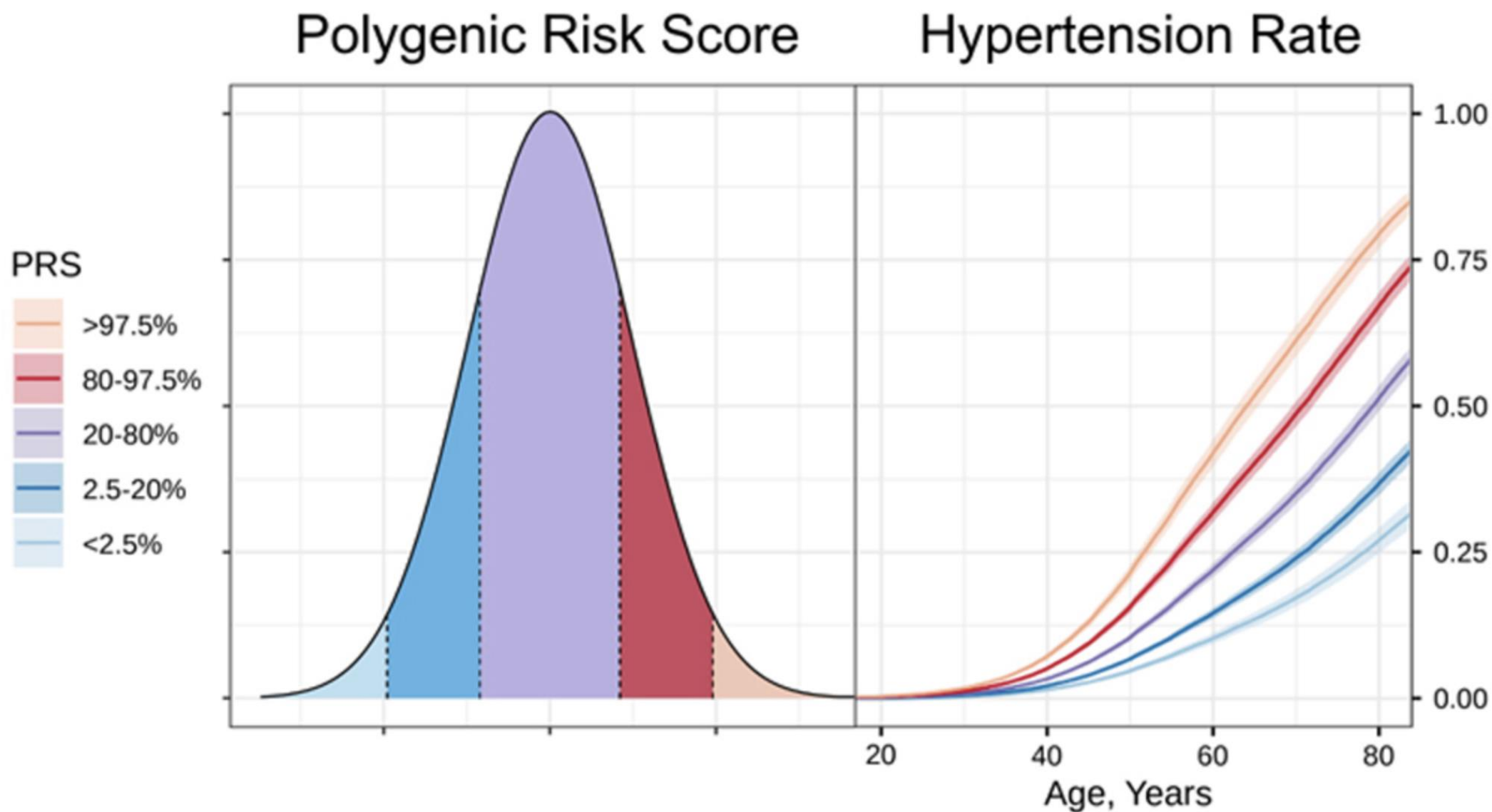
- Standardize to mean 0, SD 1 (z-score) to compare across studies
- Risk is often reported per SD or by percentile (top 20%, top 1%, etc.)
- Tail risk can look large even when average performance is modest

Standardized PRS (z-score)



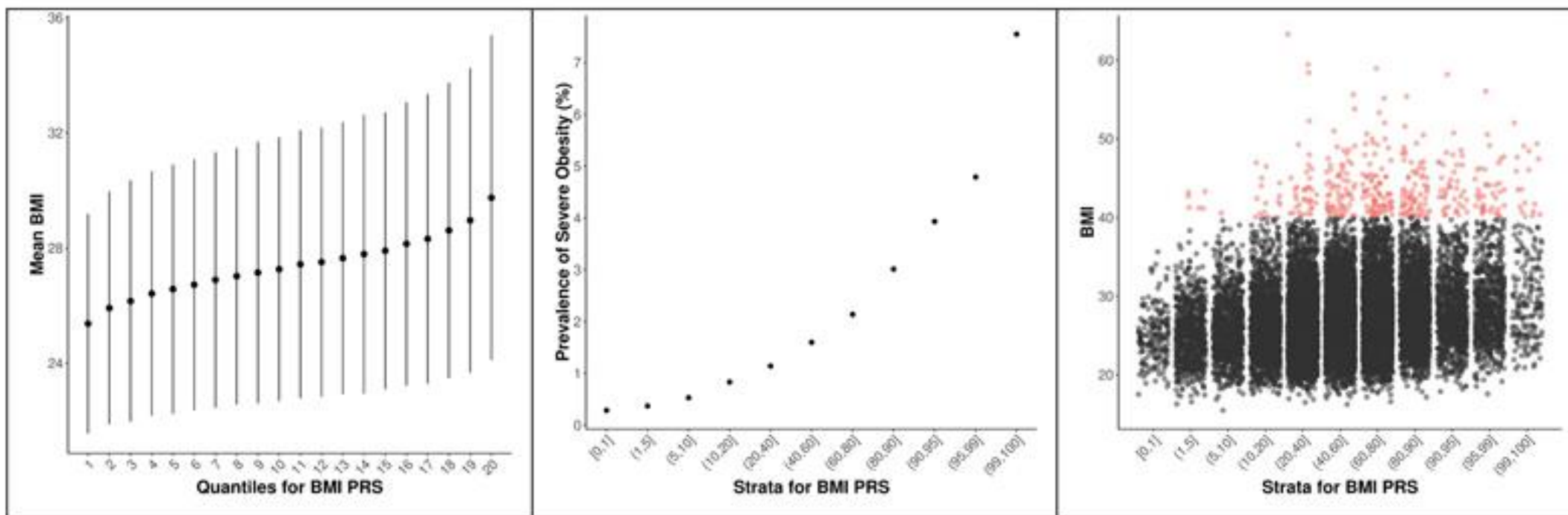
Tip: “Top 5% PRS” is a percentile within the chosen reference population.





PRS can look modest on average, but risk can concentrate in the tail

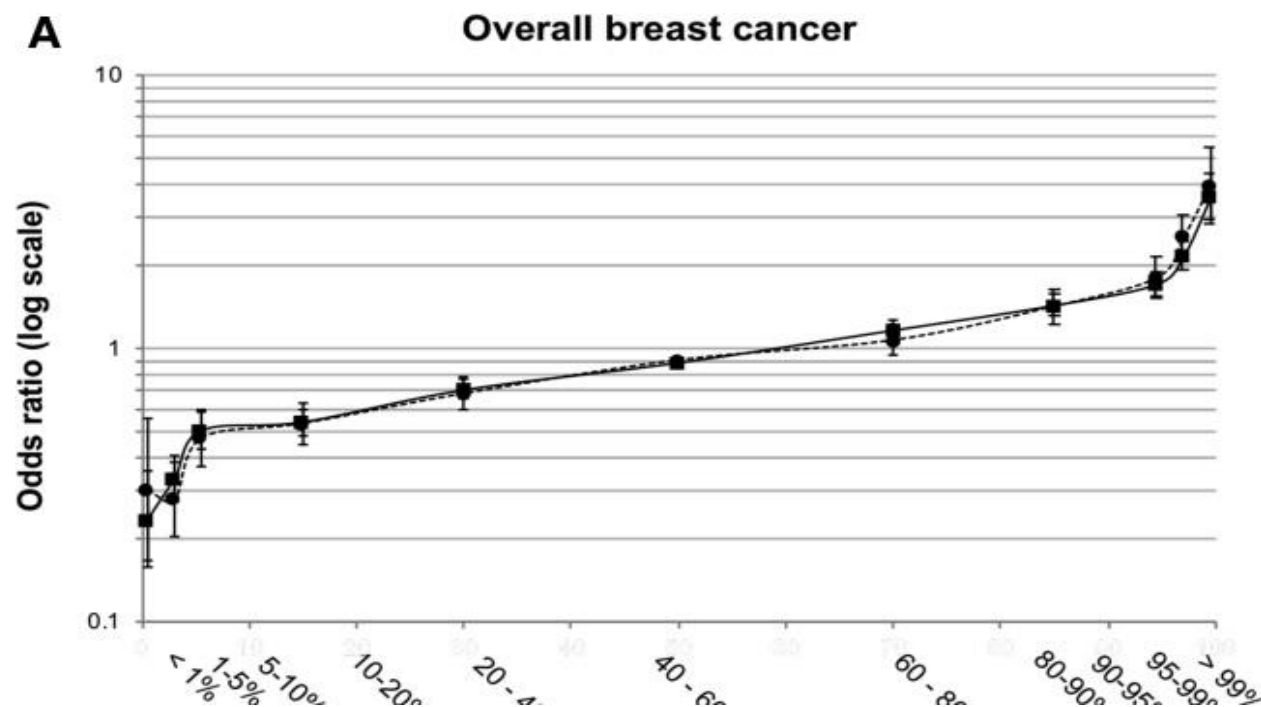
- BMI PRS in UK Biobank shown as quantiles/strata
- Mean BMI increases gradually with PRS, but severe obesity rises steeply in high-PRS strata
- Good example of tail risk and individual uncertainty



Example: breast cancer risk across PRS percentiles

Relative risk is often reported as odds ratio by percentile

- Mavaddat et al. developed and validated a 313-SNP PRS for breast cancer
- Relationship is approximately linear on the log-odds scale (polygenic model)
- Extreme percentiles can have multi-fold relative risk



Reported in this study:

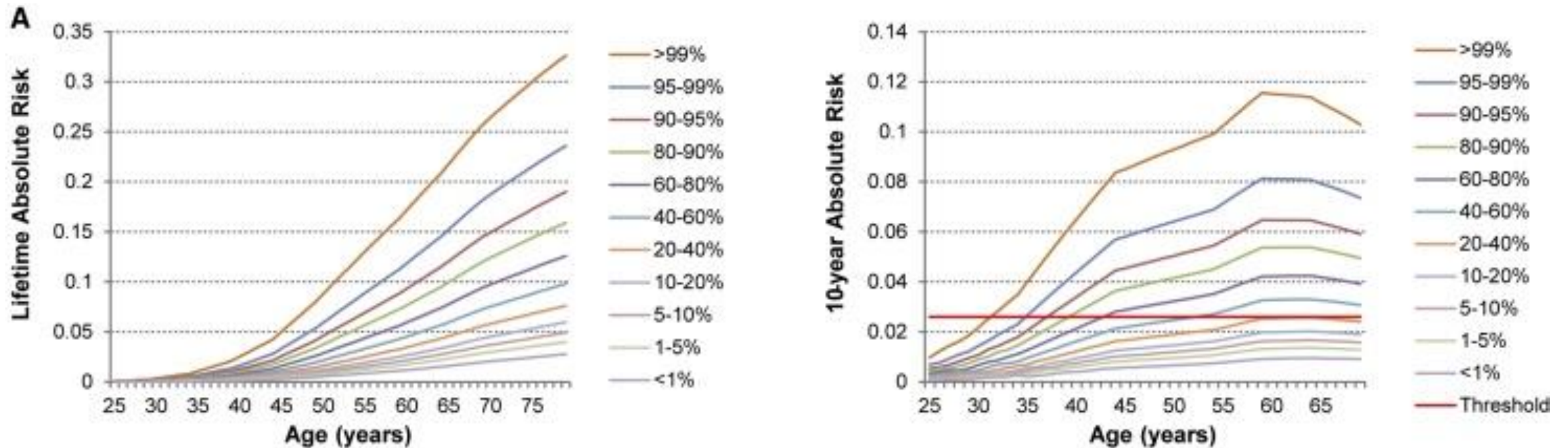
OR per 1 SD ≈ 1.61 ; AUC ≈ 0.63 (prospective test set)

Lifetime risk in top centile reported as $\sim 32.6\%$ (UK incidence model)

From relative to absolute risk: age-specific curves

Absolute risk = baseline incidence × (relative risk from PRS)

- This is what you need for screening thresholds and counseling
- Absolute risk depends on age/sex and country-specific incidence (plus competing risks)
- PRS percentiles shift the age at which you cross a given risk threshold

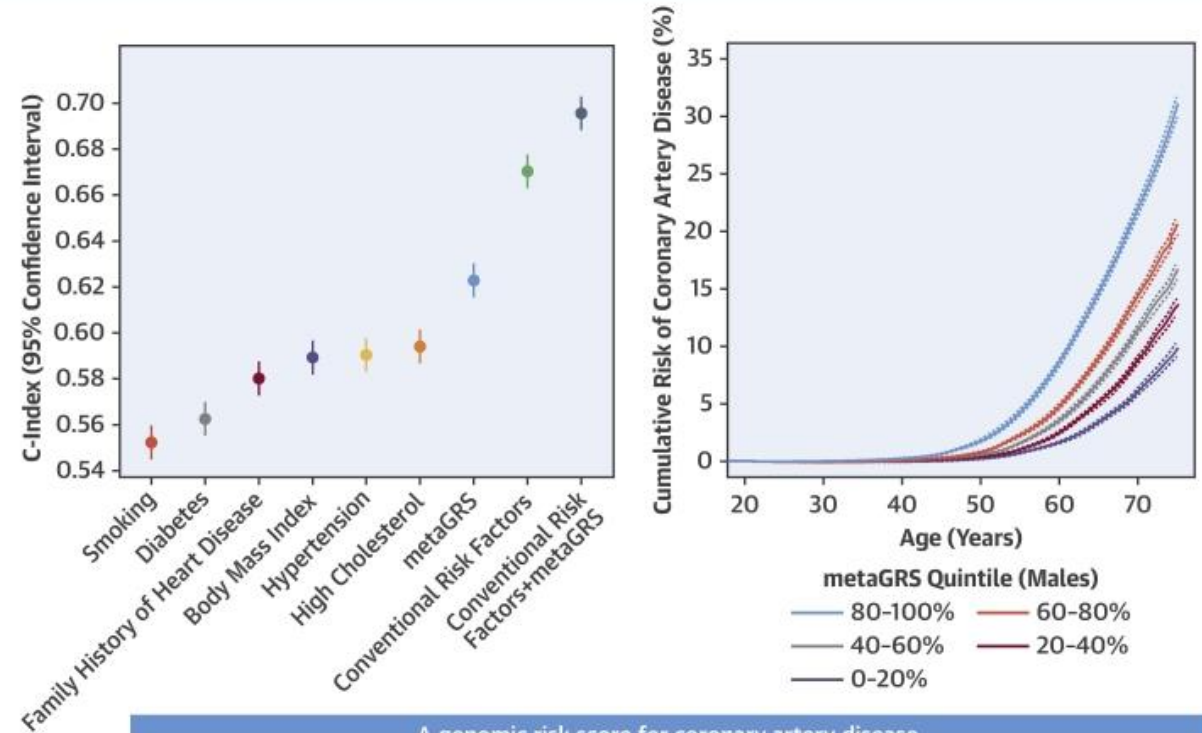


Example: coronary artery disease (CAD) genomic risk score

Large GWAS enables stronger stratification for some diseases

- Inouye et al. built a CAD score (metaGRS) from ~1.7M variants
- Top vs bottom quintile shows markedly different cumulative risk trajectories
- Genetic scores can be informative early in life, before many clinical risk factors manifest

CENTRAL ILLUSTRATION: Genomic Risk Score for Coronary Artery Disease



A genomic risk score for coronary artery disease

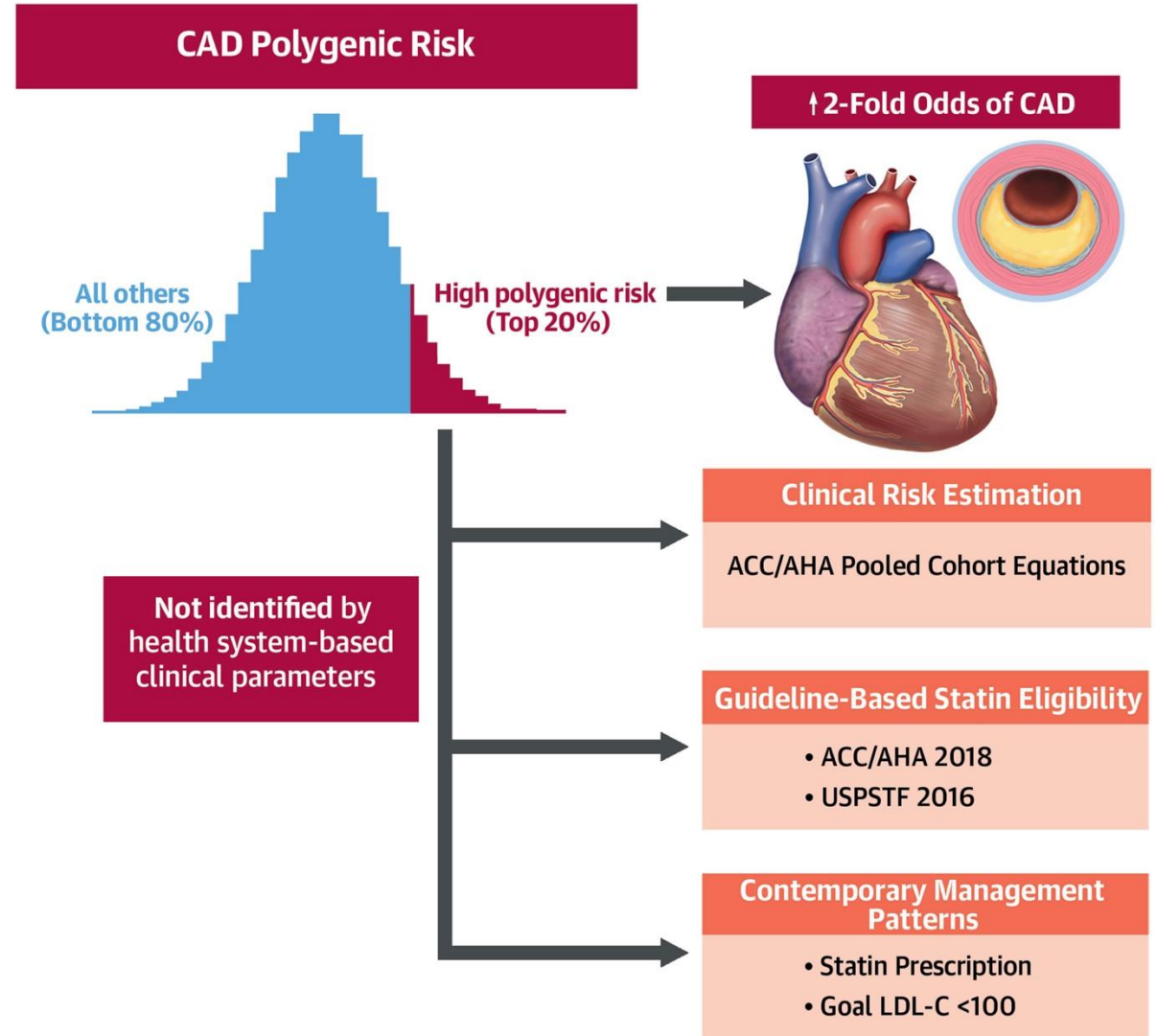
Greater association with future coronary artery disease than any single conventional risk factor
Independent of yet complements conventional risk factors
Provides meaningful lifetime risk estimates of coronary artery disease
Quantifiable at or before birth and shows potential for risk screening in early life

Inouye, M. et al. J Am Coll Cardiol. 2018;72(16):1883-93.

Example: PRS in healthcare settings (who gets missed?)

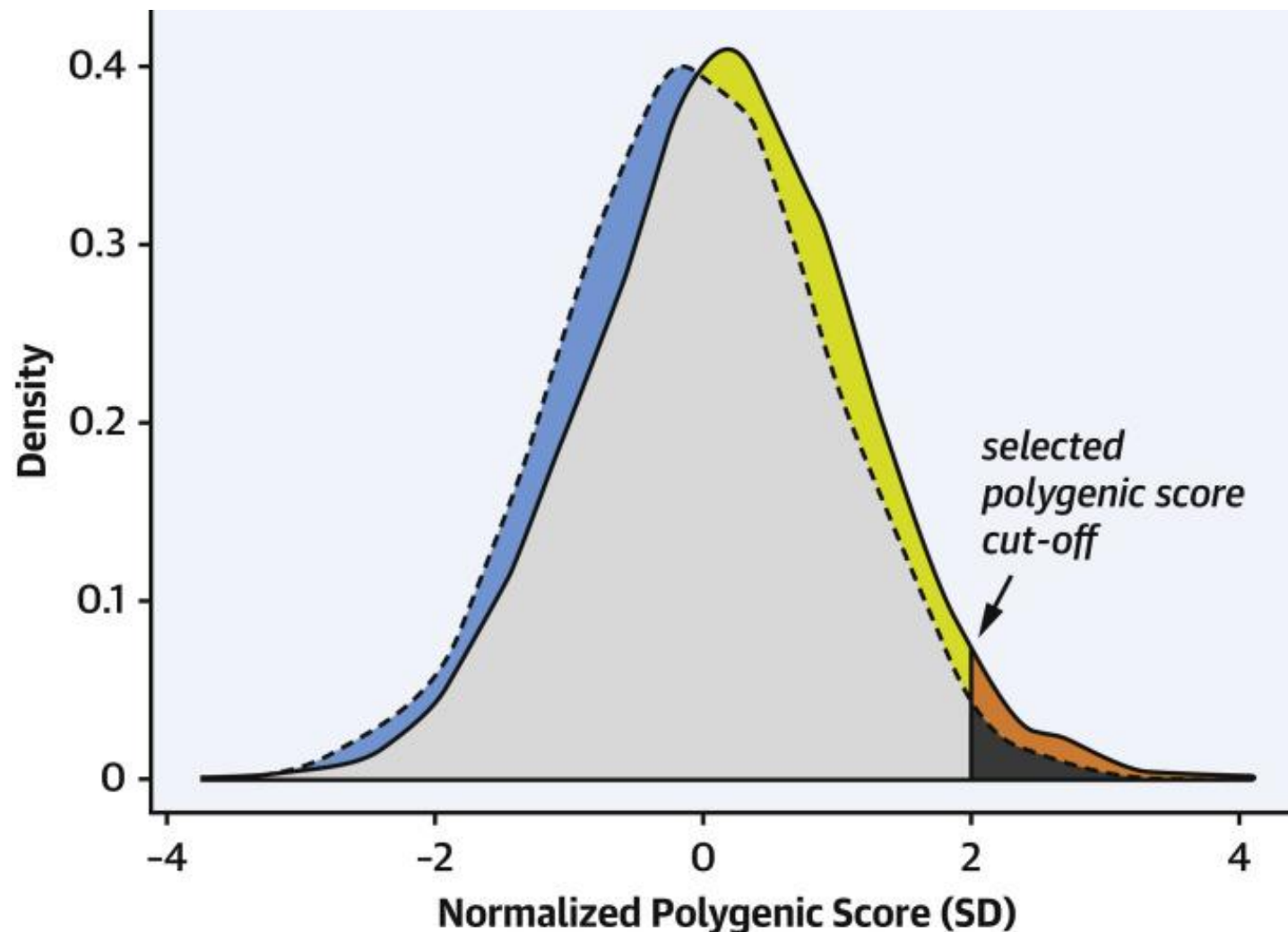
A PRS can identify people at elevated inherited risk not flagged by standard tools

- Aragam et al. applied a CAD PRS in ~47k patients across 3 U.S. healthcare systems
- Top 20% PRS had ~1.9× odds of CAD (prevalent disease) in their analysis
- Guideline-based statin eligibility and prescriptions did not distinguish high-PRS patients



Even with a strong PRS, case/control distributions overlap

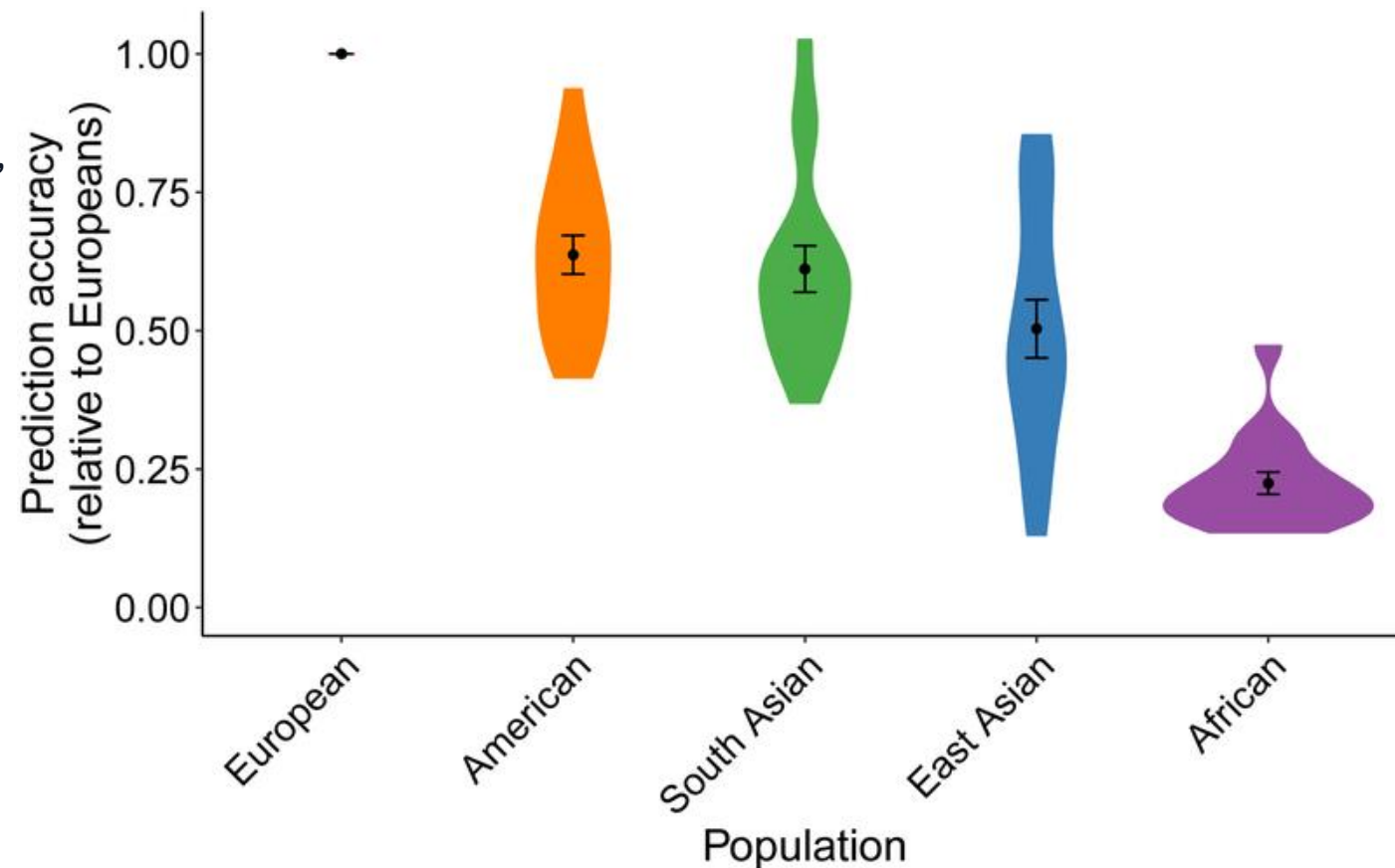
- PRS shifts probability; it does not give a definitive yes/no
- Choosing a cutoff trades false positives vs detection rate (like any screening marker)
- Utility depends on prevalence, intervention effectiveness, and harms of action



A major limitation: portability across ancestries

PRS trained in one population often performs worse in others

- Prediction accuracy decays with genetic divergence from the discovery GWAS cohort
- Drivers: allele frequency differences, LD differences, and environment/measurement differences
- Equity concern: scores can be best for Europeans if GWAS are Euro-biased



Summary

- PRS uses GWAS effect sizes to compute an individual-level genetic liability score
- Main modeling challenges: LD, shrinkage, and honest out-of-sample evaluation
- Interpretation is easiest via percentiles; decisions require absolute risk + calibration
- PRS can stratify risk for some diseases, but distributions overlap
- Portability across ancestries is a major limitation; diverse GWAS is essential